

Based on everything discussed — the SDOML v2 structure, the physics critique, the channel selection reasoning, the label noise warnings, and the pitfalls — here is the complete step-by-step data construction plan.

Phase 1 — Raw Data Access

Step 1.1 — Connect to AWS S3 anonymously

- Use `s3fs.S3FileSystem(anon=True)`
- Target bucket: `gov-nasa-hdrl-data1/contrib/fdl-sdoml/fdl-sdoml-v2/`
- Open `sdomlv2_small.zarr` (512×512) as the working store via `zarr.open_group`

Step 1.2 — Enumerate and inventory year groups

- List all digit keys from root: `["2010", "2011", ... "2020"]`
- For each year, record available channel keys and their array shapes (T, 512, 512)
- Log missing channels per year as warnings — do not silently skip
- Record T per year per channel — AIA and HMI will differ because cadence differs

Step 1.3 — Wrap all arrays as lazy Dask arrays

- `da.from_zarr(arr, chunks=(15, 512, 512))` per channel per year
 - Nothing is downloaded at this step — this is purely a computation graph
 - Build a nested dict: `lazy_store["2012"]["171A"]` → dask array
-

Phase 2 — Channel Selection

Step 2.1 — Retain the following channels only

Based on the analysis above, the working set is:

- 131A — hot flare plasma (Fe XXI, ~10 MK), earliest AIA response to flare impulsive phase
- 171A — primary coronal structure, loop evolution, filament channels, post-flare arcades
- 193A — warm corona + hot Fe XXIV secondary peak, coronal dimming detection
- 1600A — UV continuum, flare ribbons in upper chromosphere, spatial ground truth for flare location

- Bx, By, Bz — HMI vector magnetogram, all three components required for Jz and PIL computation

Step 2.2 — Explicitly exclude the following and document why

- 94A — poor quiet-sun SNR, mostly noise outside flare events; marginal gain over 131A
- 211A — largely redundant with 193A for eruption forecasting
- 304A — chromospheric contamination dominates; filament detection unreliable without H α
- 335A — weakest coronal signal in the AIA set, poor SNR, minimal literature support for flare precursors
- 1700A, 4500A — photospheric channels, minimal coronal eruption sensitivity

This gives a 7-channel working set: 4 AIA + 3 HMI.

Phase 3 — Timestamp Alignment

Step 3.1 — Load timestamp indices

- Extract timestamps from Zarr metadata `.attrs["T_OBS"]` or companion CSV per year per channel
- AIA cadence is 6 minutes; HMI cadence is 12 minutes
- Store as pandas `DatetimeIndex` per channel per year

Step 3.2 — Build unified 12-minute timeline

- Anchor on HMI timestamps — they are the coarser grid and are the limiting factor
- For each HMI timestamp `t_hmi`, find nearest AIA frame across all 4 AIA channels independently where $|t_{\text{aia}} - t_{\text{hmi}}| < 3 \text{ min}$
- If any AIA channel has no frame within the tolerance window for a given `t_hmi`, mark that timestamp as partially missing — do not silently substitute

Step 3.3 — Log coverage statistics

- Per year, report: total HMI timestamps, % successfully matched to all 4 AIA channels, % partially matched, % dropped
 - This tells you the true effective coverage before you build any windows
-

Phase 4 — Data Cleaning

Step 4.1 — Cosmic ray removal

- Per frame, per channel: apply 3×3 median filter; flag pixels where raw value deviates $> 5\sigma$ from local median
- Replace flagged pixels with local median value
- Log the fraction of pixels flagged per channel per year — a sudden spike indicates an instrument anomaly, not noise

Step 4.2 — Missing frame interpolation

- For isolated single-frame gaps within a sequence: linear interpolation between flanking frames
- For two or more consecutive missing frames: mark the entire timestamp as unusable — do not interpolate across a gap this wide
- Keep a boolean valid mask per timestamp per channel — do not drop records yet, just flag them

Step 4.3 — HMI disambiguation artifact detection

- Compute frame-to-frame difference of disk-integrated $|B_x|$ across consecutive HMI frames
- Flag timestamps where $|\Delta B_{x_integrated}|$ exceeds 3σ of the rolling 30-frame std
- These are likely 180° disambiguation jumps, not real solar evolution
- Mark flagged timestamps as unusable in the valid mask

Step 4.4 — Degradation verification

- Compute the monthly 95th-percentile intensity for each AIA channel across 2010–2020
- Plot these — they should be flat or show only solar-cycle modulation, not a long-term decline
- If a systematic downward trend is present, SDO ML v2's correction is incomplete for that channel — apply a per-year multiplicative correction factor derived from the trend residual
- HMI absolute calibration is more stable; skip this check for HMI

Step 4.5 — Solar limb mask

- Compute pixel distance from disk centre in each 512×512 frame

- Set all pixels beyond $0.98 R_{\odot}$ to zero for all channels
 - Apply consistently before any normalisation or feature extraction
 - This is critical for HMI: limb pixels have severe projection and noise artifacts
-

Phase 5 — Derived Physical Feature Maps

These are computed after cleaning, before normalisation, and stored alongside the raw channels as additional inputs. They address the physics critique directly.

Step 5.1 — Vertical current density J_z map

- From HMI B_x , B_y per frame: $J_z = (\partial B_y / \partial x - \partial B_x / \partial y) / \mu_0$
- Approximate spatial derivatives with central finite differences
- Apply the limb mask before computing — limb pixels create large spurious gradients
- Store J_z as an 8th input channel alongside the 7 raw channels
- Physical significance: high $|J_z|$ at PILs is the strongest photospheric indicator of free magnetic energy

Step 5.2 — Polarity inversion line (PIL) mask

- From HMI B_z per frame: find zero-crossing contour of B_z where $|B_{\text{total}}| > 150$ G (strong-field threshold)
- Dilate the PIL contour by 5 pixels to create a binary mask
- Compute PIL total length (number of PIL pixels) as a scalar feature per frame
- Store the binary PIL mask as a 9th input channel
- Physical significance: strong-field PIL length is the most consistently predictive single photospheric metric across published literature

Step 5.3 — Magnetic shear angle at PIL

- Along the PIL mask from Step 5.2, compute the angle between the observed horizontal field (B_x , B_y) and the potential field direction (the direction that would minimise energy)
- Approximate potential field direction as perpendicular to the PIL local tangent
- Store mean shear angle along PIL as a scalar feature per frame
- High shear angle ($> 45^\circ$) is a precursor of both flares and eruptive CMEs

Step 5.4 — Unsigned flux emergence rate

- Compute total unsigned magnetic flux per frame: $\Phi = \sum |B_z| \times \text{pixel_area}$
- Compute $\partial\Phi/\partial t$ as the finite difference across consecutive HMI frames (12-minute steps)
- Store as a scalar time series — used as input feature and as the basis for the flux continuity loss term
- Positive $\partial\Phi/\partial t$ over 6–24 hours indicates active emergence, a precursor context flag

Step 5.5 — Coronal loop gradient map from AIA 171Å

- Compute the spatial gradient magnitude $|\nabla I_{\{171\}}|$ per frame
 - This highlights loop boundaries and brightening fronts
 - Store as a 10th input channel
 - Physical significance: loop brightening in 171Å in the 6–12 hours before eruption is one of the most reliably extractable short-term AIA precursors
-

Phase 6 — Normalisation

Step 6.1 — AIA channel normalisation

- Apply arcsinh stretch: $x = \text{arcsinh}(x / a)$ where $a = \text{channel_median_train} \times 0.01$
- Median is computed on the training split only (2010–2014) — never on validation or test data
- After stretch, standardise to zero mean and unit variance using training-split statistics
- Store the per-channel (a, mean, std) triplet — these are the scalers for inference

Step 6.2 — HMI channel normalisation

- Apply tanh clip: $x = \tanh(x / B_{\text{sat}})$ where $B_{\text{sat}} = 1500$ G for B_x , B_y , B_z
- Saturates strong active region fields without losing weak field structure
- After clip, standardise using training-split mean and std

Step 6.3 — Derived feature normalisation

- Jz: standardise with training-split mean and std after clipping at $\pm 5\sigma$ (Jz has very fat tails at PILs)
- PIL mask: binary, no normalisation needed
- Shear angle: already in degrees (0–90°), normalise to [0, 1] by dividing by 90

- Flux emergence rate: standardise with training mean and std; clip at $\pm 5\sigma$
- 171Å gradient map: standardise with training mean and std

Step 6.4 — Surya scaler alignment for Models D and F

- Download scalers.yaml from [nasa-ibm-ai4science/Surya-1.0](#) via `hf_hub_download`
 - For the 4 AIA channels and 3 HMI channels that overlap with Surya's training set, replace your computed scalers with Surya's values
 - For derived channels (Jz, PIL, shear, gradient), Surya has no equivalent scalers — use your own
 - This ensures the pretrained encoder receives inputs in the distribution it was trained on
-

Phase 7 — Label Generation

Step 7.1 — Fetch GOES flare catalog

- Download NOAA GOES XRSB 1-minute flux data for 2010–2020 from NOAA FTP
- Parse flare event list: extract peak time, peak flux, start time, NOAA active region number
- Keep only M-class and X-class events (peak flux $\geq 1 \times 10^{-5}$ W/m²)
- Separate M-class and X-class labels — do not merge them into a single binary
- GOES flare catalog is more complete and more reliable than DONKI for flare events — use GOES as primary for flares, not DONKI

Step 7.2 — Fetch DONKI CME catalog

- Query NASA CCMC DONKI API for CMEs 2010–2020
- Apply speed threshold: keep only CMEs with estimated linear speed > 500 km/s
- Below this threshold treat the label as uncertain, not negative — do not assign `cme_flag = 0` to slow CMEs
- Flag slow-CME timestamps as ambiguous and exclude them from the negative class during training
- Record source active region and position angle where available

Step 7.3 — Assign labels to sliding windows

- For each window with forecast targets at [t+6h, t+12h, t+24h, t+48h], check each horizon independently:
 - flare_M_flag[h] = 1 if an M-class flare peak occurs between t and t+h
 - flare_X_flag[h] = 1 if an X-class flare peak occurs between t and t+h
 - cme_flag[h] = 1 if a fast CME (> 500 km/s) occurs between t and t+h; set to -1 (uncertain) if only slow CMEs present; 0 if nothing in DONKI
- Keep M and X labels separate throughout — conflating them masks the class difficulty

Step 7.4 — Identify and flag confined flares

- Roughly 40% of M-class flares are confined (no associated CME)
- For each M-class and X-class flare in GOES, check DONKI within ± 30 minutes for a CME association
- Add a confined_flag per flare event — a window labelled flare_M_flag=1, confined_flag=1 is a fundamentally different training example than flare_M_flag=1, confined_flag=0
- Models trained without this distinction will conflate eruptive and non-eruptive flares, which matters for SEP risk assessment

Phase 8 — Sliding Window Construction

Step 8.1 — Define window parameters

- Input length: $T_{in} = 8 \text{ frames} \times 12\text{-minute cadence} = 96 \text{ minutes of history}$
- Forecast horizons: 4 targets at t+6h (30 frames), t+12h (60), t+24h (120), t+48h (240)
- Input tensor shape per sample: (8, 10, 256, 256) — 8 timesteps, 10 channels (4 AIA + 3 HMI + Jz + PIL mask + 171Å gradient), 256×256 after downsampling

Step 8.2 — Stride and sampling

- Training stride: 1 step (12 min) — maximum coverage
- Validation stride: 4 steps (48 min) — reduce redundancy
- Test stride: 6 steps (72 min) — minimal overlap for clean evaluation

Step 8.3 — Window validity check

- For each candidate window, check the valid mask from Phase 4

- Reject window if: more than 1 frame in the input sequence is flagged invalid across any channel, or any target frame is flagged invalid
- Log total window count before and after filtering per year

Step 8.4 — Continuous sampling — do not restrict to pre-event periods

- Sample all valid windows uniformly across the full timeline — this is critical
- Do not select only windows preceding known events — this creates selection bias as described in the pitfalls section
- The label generation in Phase 7 handles the imbalance — most windows will have all-zero labels, which is correct and reflects the real prior probability

Step 8.5 — Spatial downsampling

- Downsample all channels from 512×512 to 256×256 using area averaging
- Use `skimage.transform.downscale_local_mean` or equivalent
- Area averaging preserves integrated flux — do not use bilinear or bicubic interpolation for physical channels

Phase 9 — Train / Validation / Test Split

Step 9.1 — Chronological split boundaries

Split	Period	Solar Cycle Context
Train	2010–2014	Cycle 24 rising + maximum
Validation	2015–2017	Cycle 24 declining
Test	2018–2020	Cycle 24 minimum + Cycle 25 rising

Step 9.2 — Verify no normalisation leakage

- Confirm all scaler statistics (Phase 6) were computed only on 2010–2014 frames
- Confirm Surya scalers were not re-fitted on your data at all
- Run a check: compute mean and std of each channel on validation and test sets using training scalers — report the shift; it should be modest for most channels but larger for activity-sensitive channels like 131Å around solar minimum

Step 9.3 — Verify no hyperparameter leakage

- Hyperparameter tuning (learning rate, loss weights, LoRA rank) must use validation set only

- Test set is touched once per model, for final reported numbers — not during development

Step 9.4 — Document class rates per split

- Report M-flare rate, X-flare rate, fast-CME rate separately for train, val, and test splits
 - These will differ due to Solar Cycle — this is expected, not a bug
 - These numbers are required context for interpreting evaluation metrics, especially if performance drops on the test set relative to validation
-

Phase 10 — Dataset Validation Checks

Before training, run these checks on the assembled dataset.

Step 10.1 — Label sanity

- Verify known major events are correctly labelled: X9.3 flare on 6 September 2017, X8.2 on 10 September 2017, Bastille Day flare precursors
- Spot-check 5 randomly selected positive windows per label type — visually inspect the AIA 131Å and HMI Bz frames to confirm the active region is on disk and the eruption signature is visible

Step 10.2 — Class balance audit

- Count flare_M_flag, flare_X_flag, cme_flag positive rates across full dataset
- Expected approximately: M+ rate ~1–3% of 24h windows, X-class ~0.1–0.3%, fast CME ~0.5–1.5%
- If positive rates are substantially higher, you have likely included pre-event-only sampling by mistake

Step 10.3 — Temporal autocorrelation check

- Compute autocorrelation of the flare_M_flag label series as a function of lag (in windows)
- Confirm that autocorrelation is near zero beyond a lag of ~200 windows (~40 hours)
- If high autocorrelation persists longer, your window stride is too small relative to your event duration and consecutive windows are nearly identical

Step 10.4 — Active region longitude distribution

- For positively labelled windows, compute the heliographic longitude of the source active region (from NOAA SRS daily files)

- Check that near-limb events (longitude $> 60^\circ$) are not overrepresented relative to disk-centre events
- Near-limb active regions have degraded HMI accuracy and are harder forecasting targets — their distribution in your test set affects score interpretation

Step 10.5 — Verify Jz and PIL maps visually

- For 5 known flare-productive active regions and 5 known quiet periods, plot the Jz map and PIL mask overlaid on HMI Bz
- PIL mask should align with the Bz zero-crossing in strong-field regions
- Jz should show strong localised concentrations along PILs in active regions and near-zero elsewhere
- If PIL mask is noisy or Jz is uniformly distributed, there is a bug in the derivative computation